

WSCM-Lite: Step-by-Step Mathematics

A Practitioner's Guide to the Weak Signal Cultivation Model

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Abstract

This document explains the complete mathematics of WSCM-Lite in plain language. Each step is presented as a simple procedure that can be followed with a calculator or spreadsheet. No prior knowledge of mathematics beyond basic arithmetic is required. WSCM-Lite is the practitioner-ready implementation of the Weak Signal Cultivation Model (Codourey & Gonzalez, 2026).

1 What You Are Doing

You are tracking a risk signal on a map. The map is a 10-by-10 grid with two axes:

- **x-axis (horizontal):** Risk Intensity — how bad is this signal right now?
- **y-axis (vertical):** Risk Growth Potential — how likely is it to get worse?

Every time your team meets (a “cultivation session”), you score the signal, compute its new position on the map, and record it. Over time, the signal traces a path across the map called the **risk locus**. That path tells you the story of the risk: whether it escalated, stabilized, or faded.

The map is divided into four regions at the midpoint (5, 5):

Region	Location	Meaning
Question Marks	$x < 5, y < 5$	New or unclear signal. Watch it.
Lit Fuses	$x \geq 5, y < 5$	High intensity, not yet escalating fast. Act soon.
Sleeping Cats	$x < 5, y \geq 5$	Low intensity now, but high growth potential. Don't ignore it.
Owls	$x \geq 5, y \geq 5$	High on both axes. Active management required.

Table 1: The four regions of the cultivation field.

2 Before You Start: One-Time Setup

Choose how often your team will meet. This is your **session cadence**. It sets the boundaries for classifying gaps between sessions:

That is the only configuration decision. Everything else is fixed.

3 The Seven Steps

Repeat these seven steps at every cultivation session.

Gap Type	Weekly	Biweekly	Monthly
Early ($\leq$)	5 days	10 days	22 days
Normal ($\leq$)	10 days	21 days	45 days
Missed 1 ($\leq$)	21 days	42 days	90 days
Missed 2+ ($>$)	21 days	42 days	90 days

Table 2: Gap classification boundaries by cadence. Any gap exceeding the Missed 1 threshold is classified as Missed 2+.

3.1 Step 1: Score the Signal

Each assessor independently scores the signal on two scales from 0 to 4:

- **NRS_x** (Risk Intensity): 0 = no concern, 4 = extreme concern
- **NRS_y** (Risk Growth Potential): 0 = no growth, 4 = extreme growth

Take the average of all assessors' scores and multiply by 2.5 to get the session coordinates:

$$x_{\text{new}} = 2.5 \times \text{mean}(\text{NRS}_{x,1}, \text{NRS}_{x,2}, \dots, \text{NRS}_{x,n}) \quad (1)$$

$$y_{\text{new}} = 2.5 \times \text{mean}(\text{NRS}_{y,1}, \text{NRS}_{y,2}, \dots, \text{NRS}_{y,n}) \quad (2)$$

The 2.5 multiplier maps the 0–4 NRS scale onto the 0–10 field. If three assessors all score $\text{NRS}_x = 3$, then $x_{\text{new}} = 2.5 \times 3 = 7.50$.

Entry Rule

A **new signal** may only enter the field when every individual NRS score is ≤ 1 on both axes. This means the initial position is always within $[0, 2.5] \times [0, 2.5]$ — the Question Marks region. Signals must earn their way into more serious territory through subsequent sessions.

3.2 Step 2: Classify the Gap and Look Up Weights

Count the calendar days since the last session. Use your cadence table (Section 2) to classify the gap, then look up two numbers:

Gap Type	w (recency weight)	decay
Early	0.281	0.957
Normal	0.475	0.917
Missed 1	0.700	0.840
Missed 2+	0.800	0.770

Table 3: Lookup table. These four rows replace four exponential functions from the full WSCM.

What these numbers mean:

- w (recency weight) controls how much the signal moves toward the new scores. If you missed sessions, w is higher — the new data matters more because the old position is stale.
- decay controls how much the previous y position fades. If you missed sessions, decay is lower — the old growth urgency has eroded more.

Why does y decay but x does not?

Risk Intensity (x) reflects the current state of the hazard — a broken valve stays broken whether you look at it or not. Risk Growth Potential (y) reflects urgency and momentum

— if nobody reports escalation for weeks, the sense of urgency naturally fades. The decay mechanism models this: without new evidence of growth, y drifts downward between sessions.

3.3 Step 3: Compute Committee Scaling

Adjust for how many people scored the signal:

$$c(n) = \min(1, 0.70 + 0.06 \times n) \quad (3)$$

where n is the number of assessors.

Assessors (n)	$c(n)$
1	0.76
2	0.82
3	0.88
4	0.94
5 or more	1.00

Table 4: Committee scaling values.

A solo reporter ($n = 1$) gets $c = 0.76$, which limits how far the signal can move on one person’s assessment. A full committee of five or more gets $c = 1.0$ — full repositioning authority.

3.4 Step 4: Compute the Effective Weight

Multiply the recency weight by the committee scaling:

$$w_{\text{eff}} = w \times c(n) \quad (4)$$

This is the fraction of the “distance to new data” that the signal actually moves. It ranges from 0.214 (early gap, solo reporter) to 0.800 (missed 2+ gap, full committee).

Worked Example

A biweekly team meets after 14 days (Normal gap) with 3 assessors.

- $w = 0.475$ (from the lookup table)
- $c(3) = \min(1, 0.70 + 0.06 \times 3) = \min(1, 0.88) = 0.88$
- $w_{\text{eff}} = 0.475 \times 0.88 = 0.418$

The signal will move 41.8% of the way from its current position toward the new scores.

3.5 Step 5: Update the Position

This is the core calculation. You need the signal’s previous position (x, y) and the new session coordinates ($x_{\text{new}}, y_{\text{new}}$).

Update x (Risk Intensity):

$$x' = x + w_{\text{eff}} \times (x_{\text{new}} - x) \quad (5)$$

In words: start at the old x , then move w_{eff} of the way toward the new x .

Update y (Risk Growth Potential):

First, let the old y decay:

$$y_{\text{decay}} = y \times \text{decay} \quad (6)$$

Then update from the decayed value:

$$y' = \max(0.50, y_{\text{decay}} + w_{\text{eff}} \times (y_{\text{new}} - y_{\text{decay}})) \quad (7)$$

The $\max(0.50, \dots)$ floor ensures a signal never silently disappears from the map. Even if nobody mentions it for months, y bottoms out at 0.50 — the signal stays visible in the register until the team actively retires it.

Worked Example (continued)

Previous position: (6.77, 2.86). New scores: $x_{\text{new}} = 7.50$, $y_{\text{new}} = 5.00$. Gap: 7 days (Early). $n = 3$ assessors. $w = 0.281$, decay = 0.957, $c(3) = 0.88$, $w_{\text{eff}} = 0.247$.

Update x :

$$x' = 6.77 + 0.247 \times (7.50 - 6.77) = 6.77 + 0.247 \times 0.73 = 6.77 + 0.18 = 6.95$$

Update y :

$$y_{\text{decay}} = 2.86 \times 0.957 = 2.74$$

$$y' = \max(0.50, 2.74 + 0.247 \times (5.00 - 2.74)) = \max(0.50, 2.74 + 0.56) = 3.30$$

New position: (6.95, 3.30) — still in Lit Fuses, but y is climbing.

What if no new assessment is collected?

If the signal is reviewed but not rescored in a session:

$$x' = x \quad (\text{unchanged}) \quad y' = \max(0.50, y \times \text{decay})$$

The x position stays the same. The y position decays. This means a signal that is consistently ignored will slowly drift downward on the growth axis — but never below 0.50.

3.6 Step 6: Compute Distance and Check SMS Escalation

Compute how far the signal is from the origin:

$$d = \sqrt{x'^2 + y'^2} \quad (8)$$

SMS Escalation Rule

If $d \geq 7.07$, the signal is escalated to the formal Safety Management System (SMS). The value 7.07 is the distance from the origin to the midpoint (5, 5):

$$\sqrt{5^2 + 5^2} = \sqrt{50} \approx 7.07$$

A signal that crosses this threshold has reached a level of combined intensity and growth potential that warrants formal organizational intervention. The signal remains in the WSCM register for continued tracking even after SMS escalation.

3.7 Step 7: Compute the Session Severity Index

The SSI is a reporting metric. It does not affect the signal's position.

$$S = \frac{d}{14.14} \times \ln(1 + f) \quad (9)$$

where:

- d is the distance from Step 6
- $14.14 = \sqrt{200}$ is the maximum possible distance (from the origin to the corner at (10, 10)), used for normalization
- f is the cumulative number of real-world occurrences within the current observation year
- $\ln$ is the natural logarithm

The $\ln(1 + f)$ term means early occurrences have more impact on severity than later ones. Going from $f = 0$ to $f = 5$ increases SSI substantially. Going from $f = 50$ to $f = 55$ barely changes it. This prevents the severity score from being dominated by frequency alone.

SSI Range	Status	Recommended Action
0.0 – 0.5	Low	Routine monitoring; log and observe
0.5 – 1.5	Moderate	Team review recommended
1.5 – 2.5	Elevated	Management notification; increase session frequency
≥ 2.5	Critical	Escalation imminent or SMS already triggered

Table 5: SSI interpretation bands.

4 Signal Retirement

A signal may be retired when **both** of the following are true:

1. The distance $d < 1.0$ — meaning the signal is very close to the origin, with low concern on both axes.
2. The team makes an active, deliberate decision to close it.

Signals are **never automatically retired**. The decay floor ($y_{\min} = 0.50$) ensures that y never reaches zero on its own, which means d never falls below the closure threshold without x also being very low. Retirement always requires human judgment.

The risk locus is retained permanently in the organizational record, even after retirement.

5 Complete Parameter Reference

Constant	Value	Role
n_{ref}	5	Full committee size (where $c(n) = 1.0$)
$\alpha_{\min}$	0.70	Solo reporter floor $\rightarrow c(1) = 0.76$
$y_{\min}$	0.50	Decay floor — signal never silently disappears
d_{SMS}	7.07	SMS escalation threshold
d_{close}	1.0	Minimum distance for retirement eligibility
d_{max}	14.14	$\sqrt{200}$ — maximum field distance (SSI normalization)

Table 6: All hardcoded constants. No free parameters require facilitator tuning.

6 Summary: What You Need Per Session

For each signal, at each session, you need:

1. The **day number** (calendar day from project start)
2. The **number of assessors** (n)
3. Each assessor’s **NRS scores** (0–4) for both x and y
4. The **cumulative occurrence count** (f) — updated if new real-world occurrences have been reported since the last session

Everything else is computed. The lookup table, the constants, and the formulas are all fixed. No tuning, no calibration, no software required beyond a spreadsheet.